30-Minute Student Task

Build Your Own Rule-Based Python Chatbot

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Task Overview

A simple rule-based conversational chatbot was built in Python. The chatbot interacts continuously with the user, recognizes different messages, provides appropriate responses, and allows the user to exit gracefully.

Requirements Covered

- Asks the user's name before the main conversation begins.
- Uses while True: to keep the conversation running continuously.
- Uses input() to receive messages from the user.
- Cleans the input using .lower().strip() for consistent matching.
- Recognizes at least two greetings (hello, hi).
- Answers multiple topics: how are you, bot's name, time, and help.
- Uses if / elif / else for decision making.
- Uses break to exit when the user types bye or exit.
- Uses the user's name in multiple responses.
- Provides a useful fallback response for unknown messages using else.
- Uses the in operator so variations like 'hello bot' or 'hey hello' still trigger a greeting.

Chatbot Code

```
# Rule-Based Python Chatbot
# Target flow: INPUT -> PROCESS -> RESPOND -> REPEAT -> EXIT

name = input("Hi! What's your name? ").strip()
print(f"Nice to meet you, {name}! Type 'bye' or 'exit' anytime to end our
chat.")

while True:
    message = input(f"{name}: ").lower().strip()

    if "bye" in message or "exit" in message:
        print(f"Bot: Goodbye, {name}! Have a great day.")
        break

    elif "hello" in message or "hi" in message:
        print(f"Bot: Hello {name}! How can I help you today?")

    elif "how are you" in message:
        print("Bot: I'm just a program, but I'm doing great! How about you?")

    elif "your name" in message:
        print("Bot: I'm ChatBot, your friendly rule-based assistant.")
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    elif "time" in message:
        print("Bot: Sorry, I can't check the real time yet, but I'm always
here to chat!")

    elif "help" in message:
        print(f"Bot: Sure {name}, you can ask me about my name, how I am, or
just say hi!")

    else:
        print("Bot: Sorry, I didn't understand that. Could you rephrase?")
```

Sample Test Run

Test with: hello -> Bot: Hello <name>! How can I help you today?

Test with: HELLO -> matched (input is lowercased/stripped before checking)

Test with: hi -> Bot: Hello <name>! How can I help you today?

Test with: how are you -> Bot: I'm just a program, but I'm doing great! How about you?

Test with: asdkjh (unknown message) -> Bot: Sorry, I didn't understand that. Could you rephrase?

Test with: bye or exit -> Bot: Goodbye, <name>! Have a great day. (loop ends via break)

Conclusion

The chatbot follows the INPUT -> PROCESS -> RESPOND -> REPEAT -> EXIT flow using a while True loop, cleaned string matching with the in operator, and if / elif / else branching. It personalizes responses with the user's name and exits cleanly with break when the user types bye or exit.